SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Describe the testing method of determination of permeability of a given sample of refractory.

Q.24 Describe the MgO-SiO_2 phase diagram with help of neat sketch.

Q.25 List the factors affecting of selection of refractories raw material. Describe occurance of raw material and manufacturing units in India.

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Roll No. ....

4th Sem. / Ceramic Engineering

Subject : Refractory Technology

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 The refractory is neither attacked by acid slag nor by basic slag is called

a) Acid refractory b) Basic refractory

c) Neutral refractory d) None

Q.2 PCE stands for ________.

a) Pyrometric cone equivalent

b) Pyrometric cylinder equivalent

c) Pyrometric card equivalent

d) Pyrometric care equivalent

Q.3 Silicon Carbide has chemical formula

a) CaO b) SiB

c) SiD d) SiC

Q.4 Examples of special refractory is ________

a) Dolomite refractory b) Quartz refractory

c) Zirconia refractory

d) Mag-chrom refractory

Q.5 Examples of Neutral refractory is __________.

a) Fire clay refractory

b) Silica refractory

c) Magnesite refractory

d) Carbon refractory

Q.6 _______ is the volume of air or gas which will pass through a cubic centimeter of the material under a pressure of 1 cm of water in one second.

a) Permeability b) PCE

c) CCS d) Bulk Densit

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Monolithic means single layer. (True/False)

Q.8 The main purpose of using Refractry material to retain heat in furnace. (T/F)

Q.9 Porosity of insulating refractory brick should be low. (T/F)

Q.10 Dolomite refractories are basic in nature. (T/F)

Q.11 Mullite refractories are __________ refractory.

Q.12 Silica phase diagram is on component system. (T/F)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain refractories.

Q.14 Explain preparation of chromite refractory.

Q.15 Discuss phase rule.

Q.16 Discuss mullite.

Q.17 Discuss Magnesium-Chrome refractories.

Q.18 Explain glass wool.

Q.19 Explain castable.

Q.20 List the uses of carbon refractory.

Q.21 List the classification refractories.

Q.22 Explain water absorption in refractory sample.